



## Duality

Consider the following LP:

$$\begin{array}{llll} \min & 6x_1 & +4x_2 & +2x_3 \\ \text{s.t.} & 4x_1 & +2x_2 & +x_3 \geq 5 \\ & x_1 & +x_2 & \geq 3 \\ & & x_2 & +x_3 \geq 4 \\ & & & x_i \geq 0, \text{ for } i = 1, 2, 3. \end{array}$$

Suppose you want to give a lower bound on the optimal value of this LP. What can we say?

- Since all variables are non-negative,  $6x_1 + 4x_2 + 2x_3 \geq 4x_1 + 2x_2 + x_3$ . So the value of the LP must be at least 5.
- By the same argument,  $6x_1 + 4x_2 + 2x_3 \geq (4x_1 + 2x_2 + x_3) + 2(x_1 + x_2) \geq 5 + 2 \cdot 3 = 11$ .
- Again by the same argument,  $6x_1 + 4x_2 + 2x_3 \geq (4x_1 + 2x_2 + x_3) + (x_1 + x_2) + (x_2 + x_3) \geq 5 + 3 + 4 = 12$ .



## Duality

How do we determine the best lower bound we can achieve this way? By setting up a different LP!

- Again by the same argument,  $6x_1 + 4x_2 + 2x_3 \geq (4x_1 + 2x_2 + x_3) + (x_1 + x_2) + (x_2 + x_3) \geq 5 + 3 + 4 = 12$ .

$$6x_1 + 4x_2 + 2x_3 \geq y_1(4x_1 + 2x_2 + x_3) + y_2(x_1 + x_2) + y_3(x_2 + x_3).$$

$$\text{Max } 5y_1 + 3y_2 + 4y_3$$

subject to

$$4y_1 + y_2 \leq 6$$

$$2y_1 + y_2 + y_3 \leq 4$$

$$y_2 + y_3 \leq 2$$

$$y_i \geq 0 \quad i = 1, 2, 3$$



## Duality

$$\begin{array}{llll} \min & 6x_1 & +4x_2 & +2x_3 \\ \text{s.t.} & 4x_1 & +2x_2 & +x_3 \geq 5 \\ & x_1 & +x_2 & \geq 3 \\ & & x_2 & +x_3 \geq 4 \\ & & & x_i \geq 0, \text{ for } i = 1, 2, 3. \end{array}$$

$$\begin{array}{l} \text{Min } cx \\ \text{subject to} \\ Ax \geq b \\ x \geq 0 \end{array}$$

$$\begin{array}{l} \text{Max } 5y_1 + 3y_2 + 4y_3 \\ \text{subject to} \\ 4y_1 + y_2 \leq 6 \\ 2y_1 + y_2 + y_3 \leq 4 \\ y_2 + y_3 \leq 2 \\ y_i \geq 0 \quad i = 1, 2, 3 \end{array}$$

$$\begin{array}{l} \text{Max } yb \\ \text{subject to} \\ yA \leq c \\ y \geq 0 \end{array}$$

$$\begin{array}{l} \text{Max } y^T b \\ \text{subject to} \\ y^T A \leq c \\ y \geq 0 \end{array}$$



## Duality

- Construction of the dual LP from the primal LP

| Max LP       |                   | Min LP       |
|--------------|-------------------|--------------|
| Constraints  |                   | Variables    |
| $\geq$       | $\leftrightarrow$ | $\leq 0$     |
| $\leq$       | $\leftrightarrow$ | $\geq 0$     |
| $=$          | $\leftrightarrow$ | unrestricted |
| Variables    |                   | Constraints  |
| $\geq 0$     | $\leftrightarrow$ | $\geq$       |
| $\leq 0$     | $\leftrightarrow$ | $\leq$       |
| unrestricted | $\leftrightarrow$ | $=$          |



## Weak and Strong Duality

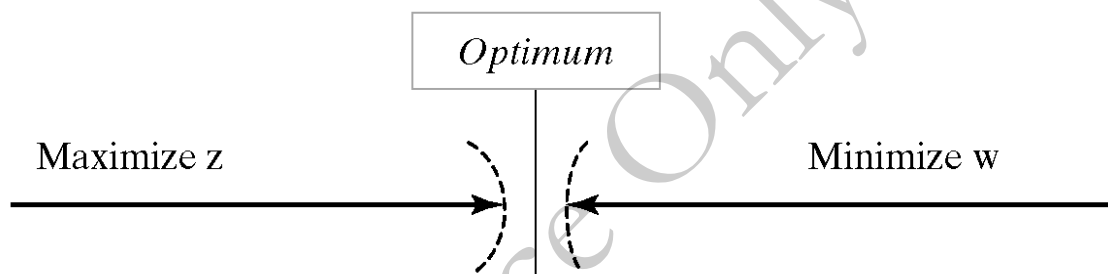


Figure 4.2

Relationship between maximum  $z$  and minimum  $w$ .



## Duality

Fundamental theorem of Duality – Only one of the statements is true.

- Both primal and dual have optimal solutions.
- One problem has an unbounded solution, while the other problem is infeasible.
- Both problems are infeasible.

|    | Primal     | then              | Dual                    |
|----|------------|-------------------|-------------------------|
| If | Optimal    | $\leftrightarrow$ | Optimal                 |
|    | Unbounded  | $\rightarrow$     | Infeasible              |
|    | Infeasible | $\rightarrow$     | Infeasible or Unbounded |

|    | Dual       | then              | Primal                  |
|----|------------|-------------------|-------------------------|
| If | Optimal    | $\leftrightarrow$ | Optimal                 |
|    | Unbounded  | $\rightarrow$     | Infeasible              |
|    | Infeasible | $\rightarrow$     | Infeasible or Unbounded |



## Duality

Prima solves the following LP (LP1) and concludes it has an optimal solution.

$$\begin{array}{ll}\text{Max} & cx \\ \text{subject to} & \\ & Ax = b \\ & x \geq 0\end{array}$$

Jewel solves the LP below (LP2) and claims it is unbounded. Assuming Prima's conclusion is correct, is Jewel's claim correct?

$$\begin{array}{ll}\text{Max} & cx \\ \text{subject to} & \\ & Ax = d \\ & x \geq 0\end{array}$$



# Simplex Tableau Structure

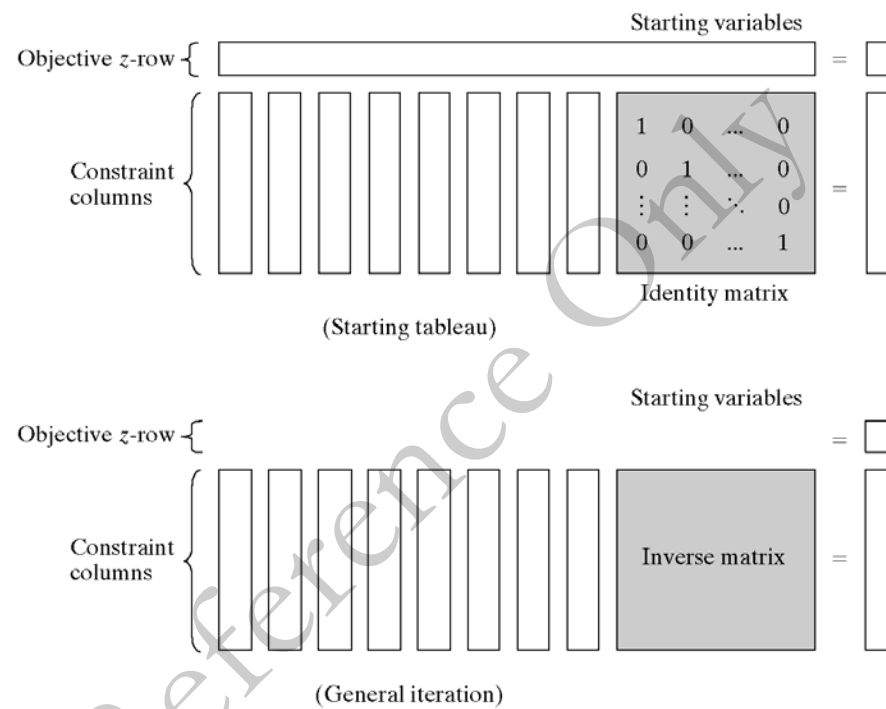


Figure 4.1  
Schematic representation of the starting and general simplex tableaus.



## Tableau Method

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

|    | x1 | x2 | s1 | s2 | s3 | s4 | Solution |
|----|----|----|----|----|----|----|----------|
| z  | -5 | -4 | 0  | 0  | 0  | 0  | 0        |
| s1 | 6  | 4  | 1  | 0  | 0  | 0  | 24       |
| s2 | 1  | 2  | 0  | 1  | 0  | 0  | 6        |
| s3 | -1 | 1  | 0  | 0  | 1  | 0  | 1        |
| s4 | 0  | 1  | 0  | 0  | 0  | 1  | 2        |



## Revisiting Reddy Mikks Problem

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

|    | x1 | x2 | s1 | s2 | s3 | s4 | Solution |
|----|----|----|----|----|----|----|----------|
| z  | -5 | -4 | 0  | 0  | 0  | 0  | 0        |
| s1 | 6  | 4  | 1  | 0  | 0  | 0  | 24       |
| s2 | 1  | 2  | 0  | 1  | 0  | 0  | 6        |
| s3 | -1 | 1  | 0  | 0  | 1  | 0  | 1        |
| s4 | 0  | 1  | 0  | 0  | 0  | 1  | 2        |



## Revisiting Reddy Mikks Problem

|    | x1 | x2   | s1   | s2 | s3 | s4 | Solution |
|----|----|------|------|----|----|----|----------|
| z  | 0  | -2/3 | 5/6  | 0  | 0  | 0  | 20       |
| x1 | 1  | 2/3  | 1/6  | 0  | 0  | 0  | 4        |
| s2 | 0  | 4/3  | -1/6 | 1  | 0  | 0  | 2        |
| s3 | 0  | 5/3  | 1/6  | 0  | 1  | 0  | 5        |
| s4 | 0  | 1    | 0    | 0  | 0  | 1  | 2        |



## Revisiting Reddy Mikks Problem

|    | x1 | x2 | s1   | s2   | s3 | s4 | Solution |
|----|----|----|------|------|----|----|----------|
| z  | 0  | 0  | 3/4  | 1/2  | 0  | 0  | 21       |
| x1 | 1  | 0  | 1/4  | -1/2 | 0  | 0  | 3        |
| x2 | 0  | 1  | -1/8 | 3/4  | 0  | 0  | 3/2      |
| s3 | 0  | 0  | 3/8  | -5/4 | 1  | 0  | 5/2      |
| s4 | 0  | 0  | 1/8  | -3/4 | 0  | 1  | 1/2      |

Maximize  $z = 5x_1 + 4x_2$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

Min  $24y_1 + 6y_2 + y_3 + 2y_4$   
subject to

$$6y_1 + y_2 - y_3 \geq 5$$

$$4y_1 + 2y_2 + y_3 + y_4 \geq 4$$

$$y_1, y_2, y_3, y_4 \geq 0$$

$$y_1 = .75, y_2 = .5, y_3 = y_4 = 0$$



## Interpreting the Dual Variables

|    | x1 | x2 | s1   | s2   | s3 | s4 | Solution |
|----|----|----|------|------|----|----|----------|
| z  | 0  | 0  | 3/4  | 1/2  | 0  | 0  | 21       |
| x1 | 1  | 0  | 1/4  | -1/2 | 0  | 0  | 3        |
| x2 | 0  | 1  | -1/8 | 3/4  | 0  | 0  | 3/2      |
| s3 | 0  | 0  | 3/8  | -5/4 | 1  | 0  | 5/2      |
| s4 | 0  | 0  | 1/8  | -3/4 | 0  | 1  | 1/2      |

Maximize  $z = 5x_1 + 4x_2$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

$$z < w$$

Profit < Worth of Resources

Min  $24y_1 + 6y_2 + y_3 + 2y_4$

subject to

$$6y_1 + y_2 - y_3 \geq 5$$

$$4y_1 + 2y_2 + y_3 + y_4 \geq 4$$

$$y_1, y_2, y_3, y_4 \geq 0$$

$$y_1 = .75, y_2 = .5, y_3 = y_4 = 0$$



## Complementary Slackness

Let  $x^*$  and  $y^*$  be any feasible solutions to the primal and dual problems respectively in the canonical form. They are optimal if and only if (iff):

- Every binding constraint in the dual has its corresponding primal variable positive.
- Every binding constraint in the primal has its corresponding dual variable positive.
- Every constraint that is not binding has its corresponding dual variable equal to zero.

$$\text{Max } z = cx$$

subject to

$$Ax \leq b$$

$$x \geq 0$$

$$\text{Min } w = yb$$

subject to

$$yA \geq c$$

$$y \geq 0$$

$$cx \leq yAx \leq yb \quad cx^* = y^*Ax^* = y^*b$$



## Complementary Slackness

$$\begin{aligned} \text{Min} \quad & 2x_1 + 3x_2 + 5x_3 + 2x_4 + 3x_5 \\ \text{subject to} \quad & x_1 + x_2 + 2x_3 + x_4 + 3x_5 \geq 4 \\ & 2x_1 - 2x_2 + 3x_3 + x_4 + x_5 \geq 3 \\ & x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

Let  $x^*$  and  $y^*$  be any feasible solutions to the primal and dual problems respectively in the canonical form. They are optimal if and only if:

- Every binding constraint in the dual has its corresponding primal variable positive.
- Every binding constraint in the primal has its corresponding dual variable positive.
- Every constraint that is not binding has its corresponding dual variable equal to zero.



## Complementary Slackness

$$\text{Min } z = 2x_1 + 3x_2 + 5x_3 + 2x_4 + 3x_5$$

subject to

$$x_1 + x_2 + 2x_3 + x_4 + 3x_5 \geq 4$$

$$2x_1 - 2x_2 + 3x_3 + x_4 + x_5 \geq 3$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

$$x_1^* = 1, x_5^* = 1, x_2^* = x_3^* = x_4^* = 0$$

$$\text{Max } w = 4y_1 + 3y_2$$

subject to

$$y_1 + 2y_2 \leq 2$$

$$y_1 - y_2 \leq 3$$

$$2y_1 + 3y_2 \leq 5$$

$$y_1 + y_2 \leq 2$$

$$3y_1 + y_2 \leq 3$$

$$y_1, y_2 \geq 0$$

$$y_1^* = 4/5, y_2^* = 3/5$$

Let  $x^*$  and  $y^*$  be any feasible solutions to the primal and dual problems respectively in the canonical form. They are optimal if and only if:

- Every binding constraint in the dual has its corresponding primal variable positive.
- Every binding constraint in the primal has its corresponding dual variable positive.
- Every constraint that is not binding has its corresponding dual variable equal to zero.

Let  $x^*$  and  $y^*$  be any feasible solutions to the primal (Min) and dual problems (Max) respectively in the canonical form. They are optimal if and only if:

- $(e_i)^*(y_i) = 0, i=1, \dots, m$
- $(s_j)^*(x_j) = 0, j=1, \dots, n$



## Dual Simplex Algorithm

- Solves the dual problem directly on the (primal) simplex tableau.
- Each iteration involves a DBFS, dual basic feasible solution.
- What is a DBFS?
- Test for Feasibility
- Min ratio test for the Entering Variable.

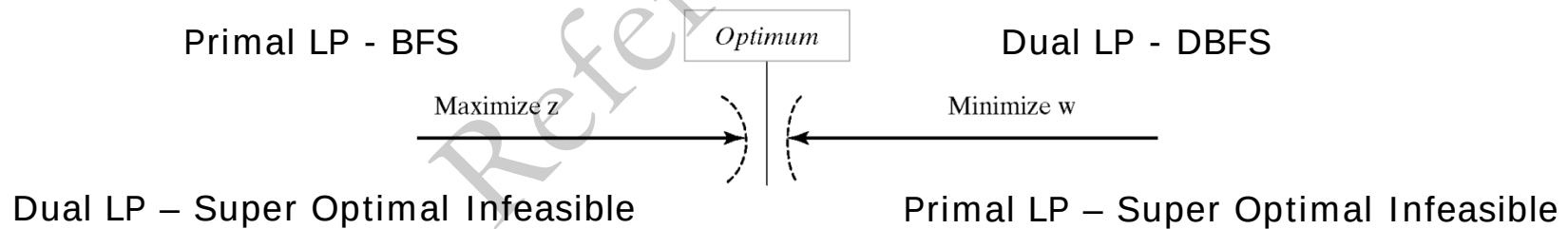


Figure 4.2

Relationship between maximum  $z$  and minimum  $w$ .



## Dual Simplex Algorithm

1. The LP is infeasible but optimal, actually super optimal.
2. Get a starting DBFS.
3. Test for Feasibility.
4. If feasible, STOP, else go to step 5.
5. Determine the Leaving Variable,  $x_r$ .
6. Determine the Entering Variable,  $x_j$ . Check for unboundedness. If unbounded, the primal is infeasible.
7. Determine the new DBFS.
8. Go to step 3.

$$\min_{\text{nonbasic } x_j} \left\{ \left| \frac{\bar{c}_j}{\alpha_{rj}} \right|, \alpha_{rj} < 0 \right\}$$



## Dual Simplex Algorithm

$$\text{Min } 3x_1 + 2x_2 + x_3$$

subject to

$$3x_1 + x_2 + x_3 \geq 3$$

$$-3x_1 + 3x_2 + x_3 \geq 6$$

$$x_1 + x_2 + x_3 \leq 3$$

$$x_1, x_2, x_3 \geq 0$$

|    | x1 | x2 | x3 | x4 | x5 | x6 | Solution |
|----|----|----|----|----|----|----|----------|
| z  | -3 | -2 | -1 | 0  | 0  | 0  | 0        |
| X4 | -3 | -1 | -1 | 1  | 0  | 0  | -3       |
| X5 | 3  | -3 | -1 | 0  | 1  | 0  | -6       |
| X6 | 1  | 1  | 1  | 0  | 0  | 1  | 3        |



## Dual Simplex Algorithm

|    | x1 | x2 | x3 | x4 | x5 | x6 | Solution |
|----|----|----|----|----|----|----|----------|
| z  | -3 | -2 | -1 | 0  | 0  | 0  | 0        |
| X4 | -3 | -1 | -1 | 1  | 0  | 0  | -3       |
| X5 | 3  | -3 | -1 | 0  | 1  | 0  | -6       |
| X6 | 1  | 1  | 1  | 0  | 0  | 1  | 3        |

$$\min_{\text{nonbasic } x_j} \left\{ \left| \frac{\bar{c}_j}{\alpha_{rj}} \right|, \alpha_{rj} < 0 \right\}$$

|    | x1 | x2 | x3   | x4 | x5   | x6 | Solution |
|----|----|----|------|----|------|----|----------|
| z  | -5 | 0  | -1/3 | 0  | -2/3 | 0  | 4        |
| X4 | -4 | 0  | -2/3 | 1  | -1/3 | 0  | -1       |
| X2 | -1 | 1  | 1/3  | 0  | -1/3 | 0  | 2        |
| X6 | 2  | 0  | 2/3  | 0  | 1/3  | 1  | 1        |



## Dual Simplex Algorithm

|    | x1 | x2 | x3   | x4 | x5   | x6 | Solution |
|----|----|----|------|----|------|----|----------|
| z  | -5 | 0  | -1/3 | 0  | -2/3 | 0  | 4        |
| X4 | -4 | 0  | -2/3 | 1  | -1/3 | 0  | -1       |
| X2 | -1 | 1  | 1/3  | 0  | -1/3 | 0  | 2        |
| X6 | 2  | 0  | 2/3  | 0  | 1/3  | 1  | 1        |

$$\min_{\text{nonbasic } x_j} \left\{ \left| \frac{\bar{c}_j}{\alpha_{rj}} \right|, \alpha_{rj} < 0 \right\}$$

|    | x1 | x2 | x3 | x4   | x5   | x6 | Solution |
|----|----|----|----|------|------|----|----------|
| z  | -3 | 0  | 0  | -1/2 | -1/2 | 0  | 9/2      |
| x3 | 6  | 0  | 1  | -3/2 | 1/2  | 0  | 3/2      |
| X2 | -3 | 1  | 0  | 1/2  | -1/2 | 0  | 3/2      |
| X6 | -2 | 0  | 0  | 1    | 0    | 1  | 0        |



## Revisiting Reddy Mikks Tableau

|    | x1 | x2 | s1   | s2   | s3 | s4 | Solution |
|----|----|----|------|------|----|----|----------|
| z  | 0  | 0  | 3/4  | 1/2  | 0  | 0  | 21       |
| x1 | 1  | 0  | 1/4  | -1/2 | 0  | 0  | 3        |
| x2 | 0  | 1  | -1/8 | 3/4  | 0  | 0  | 3/2      |
| s3 | 0  | 0  | 3/8  | -5/4 | 1  | 0  | 5/2      |
| s4 | 0  | 0  | 1/8  | -3/4 | 0  | 1  | 1/2      |

Maximize  $z = 5x_1 + 4x_2$

$$6x_1 + 4x_2 \leq 24$$

$$x_1 + 2x_2 \leq 6$$

$$-x_1 + x_2 \leq 1$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

Min  $24y_1 + 6y_2 + y_3 + 2y_4$   
subject to

$$6y_1 + y_2 - y_3 \geq 5$$

$$4y_1 + 2y_2 + y_3 + y_4 \geq 4$$

$$y_1, y_2, y_3, y_4 \geq 0$$

$$y_1 = .75, y_2 = .5, y_3 = y_4 = 0$$



# Simplex Tableau Structure

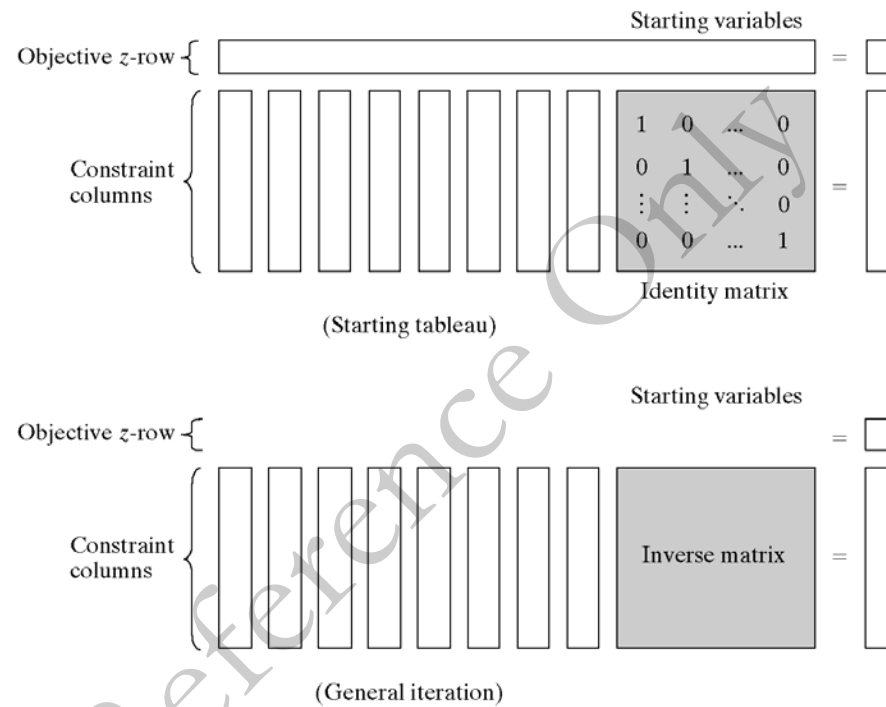


Figure 4.1  
Schematic representation of the starting and general simplex tableaus.



## Simplex Tableau Formulas

Max  $cx$   
subject to  
 $Ax \leq b$   
 $x \geq 0$

$Ax + Is = b$   
 $(x, s) = (x_b, x_n)$   
 $c = (c_b, c_n)$   
 $(A, I) = (B, N)$

Max  $z = c_b x_b + c_n x_n$   
subject to  
 $Bx_b + Nx_n = b$   
 $x_b, x_n \geq 0$

$Bx_b + Nx_n = b$   
 $x_n = 0$   
 $x_b = B^{-1}b$

$z = c_b x_b + c_n x_n$   
 $z = c_b B^{-1}b$



## Simplex Tableau Formulas

$$\text{Max } z = c_b x_b + c_n x_n$$

subject to

$$Bx_b + Nx_n = b$$

$$x_b, x_n \geq 0$$

$$Bx_b + Nx_n = b$$

$$B^{-1}Bx_b + B^{-1}Nx_n = B^{-1}b$$

$$x_b = B^{-1}b - B^{-1}Nx_n$$

$$z = c_b x_b + c_n x_n = c_b B^{-1}b - c_b B^{-1}Nx_n + c_n x_n$$

$$z = c_b B^{-1}b - c_b B^{-1}Nx_n + c_n x_n$$

$$z = -(c_b B^{-1}N - c_n)x_n$$

$$(c_b B^{-1}N - c_n) < 0$$



## Simplex Tableau Formulas

Max  $cx$   
subject to  
 $Ax \leq b$   
 $x \geq 0$

$$Ax \leq b$$

$$Ax + Is = b$$

$$B^{-1}A + B^{-1}s = B^{-1}b$$



# Simplex Tableau Structure

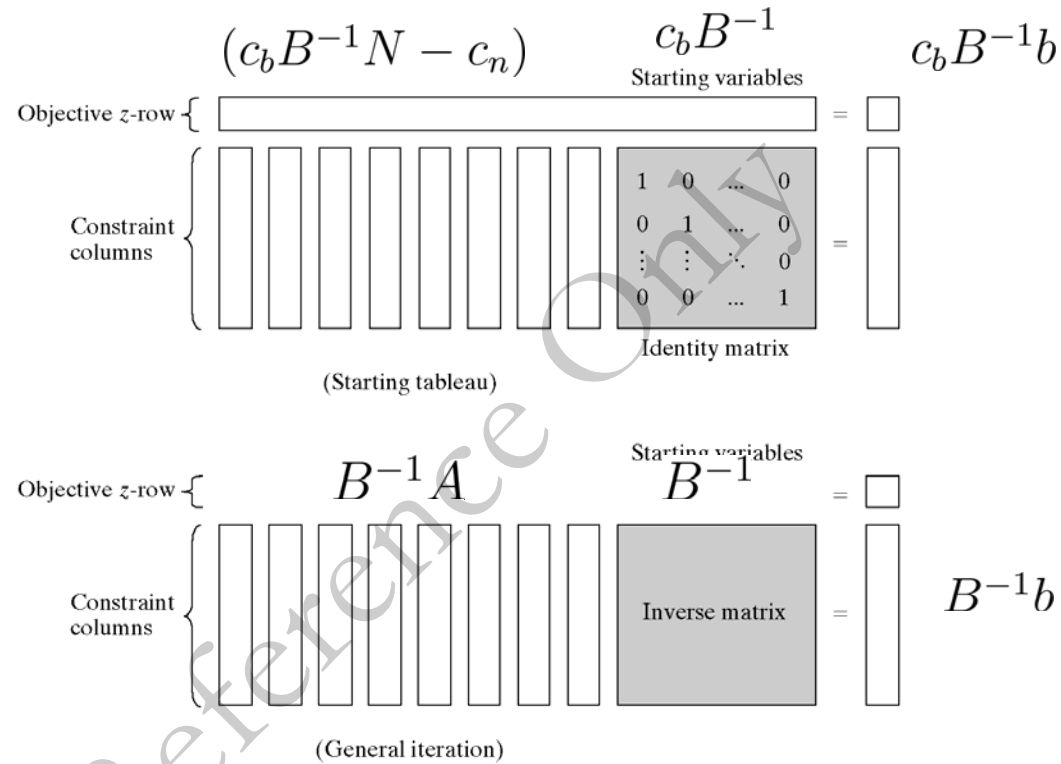


Figure 4.1

Schematic representation of the starting and general simplex tableaus.



## Simplex Tableau Formulas

|                        |              |                |
|------------------------|--------------|----------------|
| $c_b B^{-1} A_j - c_j$ | $c_b B^{-1}$ | $c_b B^{-1} b$ |
| $B^{-1} A$             | $B^{-1}$     | $B^{-1} b$     |

Max  $2x_1 + 3x_2 + 6x_3$   
 subject to  
 $x_1 + 2x_2 + 3x_3 \leq 10$   
 $x_1 + 2x_2 + 2x_3 \leq 6$   
 $x_1, x_2, x_3 \geq 0$

|    | x1    | x2 | x3 | x4  | x5    | Solution |
|----|-------|----|----|-----|-------|----------|
| Z  | 1/10  | 0  | 0  | 9/5 | 3/10  | 99/5     |
| X2 | -1/10 | 1  | 0  | 1/5 | -3/10 | 1/5      |
| X3 | 2/5   | 0  | 1  | 1/5 | 1/5   | 16/5     |